Circle and Analytical Geometry

AMC 10

Useful Formulations

- **Basics:** Fundamentally, a circle is defined as a locus of points that are equidistant or equally distanced from a central point. The area of a circle is equal to πr^2, its circumference is 2π. Along a similar vein of computation, we can find that the surface area of a sphere is $4\pi^2$ and that its volume is $\frac{4}{3}\pi r^3$.

- **Important Lines:** A *chord* is a line segment with endpoints located at the two points on the circle, a *secant* is a line that intersects the circle at two points and can extend beyond, and a *tangent* intersects the circle at exclusively one point.

- **Coordinate Plane:** Suppose that the center is (h, k) and the radius is r. If a point belongs to the circle, then it must satisfy the distance formula as shown below,

$$\sqrt{(x-h)^2 + (y-k)^2} = r$$
$$(x-h)^2 + (y-k)^2 = r^2$$

  Note that $\sqrt{(x-h)^2 + (y-k)^2}$ symbolizes the distance between (h, k) and (x, y), which must be equal to r.

- **Sectors:** The length of a circle sector, which resembles a pumpkin pie wedge, that subtends a central angle of θ degrees in a circle of radius r is equal to

$$L = \frac{\theta}{360} * 2\pi r = \frac{\theta}{180}\pi r.$$

  To grasp this intuitively, we are essentially considering a certain fraction $\frac{\theta}{360}$ of the entire circle circumference. Along a similar sequence of thought, the area of this sector is equal to

$$A = \frac{\theta}{360} * \pi r^2.$$

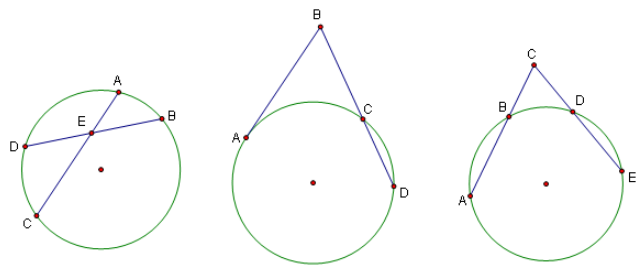
- **Angle Chasing:** Central angles, which consist of a vertex at the center of the circle, have an angle measure equal to the arc measure. Meanwhile, inscribed angles, where a vertex is positioned on the circumference, have an angle measure that is half of the arc measure. By consequence, an angle that inscribes a semicircle has an angle measure of 90 degrees, in concordance with Thales Theorem.

- **Angle Chasing II:** A few special cases concern angle measures with secants and chords. If the vertex lies external to the circle, then its angle measure is one half of the difference between the two arc measures it subtends. If the vertex is inside the circle (the junction of two chords), then its angle measure is one half of the sum, or the average, between the two arcs it subtends.

- **Power of a Point:** Let P be a point in the interior of a circle. Consider two chords passing through P, the first of which intersects the circle at A and B and the second of which intersects the circle at C and D. Then,

$$AP * BP = CP * DP$$

This can be proven expeditiously using similar triangles and angle-chasing. There are additional cases of the power of a point, with slightly modified formula. For instance, if B is a point outside of the circle and A, C, and D are distinct points on the circle such that BA is tangent to the circle that B, C, and D are collinear, then $AB^2 = BC * BD = BC(BC + CD)$.



Finally, if we consider a point C outside the circle as well as distinct points B, A, D, and E such that C, B, and A are collinear, and C, D, and E are collinear, then $CB * CA = CD * CE$. Equivalently,

$$CB(CB + BA) = CD(CD + DE)$$

- **Ellipses:** An ellipse in the coordinate plane can be defined using the following equation.

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1.$$

Specifically, this articulates an ellipse with center (h, k), a horizontal axis length of $2a$ and a vertical axis length of $2b$. Note that the area of an ellipse is πab (the area becomes $\pi * a^2$ if $a = b$, where we essentially have a circle).

This formula arises from the core definition of an ellipse, which is a locus of points where the sum of its distances to two points (called the *foci*) is precisely the same. If $a > b$, the foci are located at $(h + \sqrt{b^2 - a^2}, k)$ and $(h - \sqrt{b^2 - a^2}, k)$. Otherwise, the foci are placed vertically at $(h, k - \sqrt{b^2 - a^2})$ and $(h, k + \sqrt{b^2 - a^2})$.

- **Circle-Circle Tangency:** Among the interactions between multiple circles anchored in the coordinate plane, we can have tangency, where the circles intersect at exclusively one point, which is placed on the line between the two centers. We call the circles externally tangent if the circles are outside one another and internally tangent if one circle fully encompasses the other.

- **Ptolemy's Theorem** Let $ABCD$ be a cyclic quadrilateral (all four vertices lie on the circle's circumference). Then,

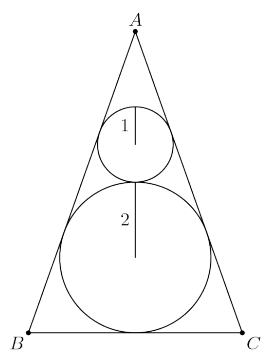
$$AB * CD + BC * AD = AC * BD.$$

In other words, the sum of the products of opposing sides is equal to the product of the diagonals.

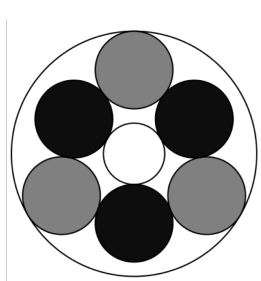
Exercises

1. Two circles intersect at points A and B. The minor arcs AB measure $30°$ on one circle and $60°$ on the other circle. What is the ratio of the area of the larger circle to the area of the smaller circle? (AMC 12A 2014)

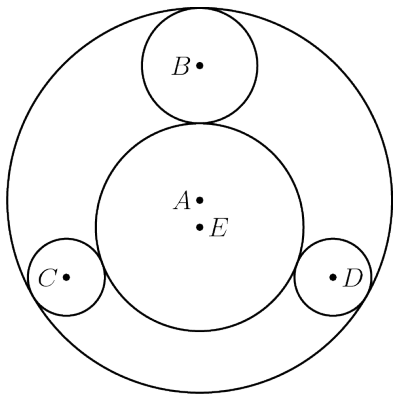
2. A circle of radius 1 is tangent to a circle of radius 2. The sides of $\triangle ABC$ are tangent to the circles as shown, and the sides $\overline{AB}$ and $\overline{AC}$ are congruent. What is the area of $\triangle ABC$? (AMC 10A 2006)



3. Two concentric circles have radii 1 and 4. Six congruent circles form a ring where each of the six circles is tangent to the two circles adjacent to it as shown. The three lightly shaded circles are internally tangent to the circle with radius 4 while the three darkly shaded circles are externally tangent to the circle with radius 1. The radius of the six congruent circles can be written $\frac{k+\sqrt{m}}{n}$ where k, m, and n are integers with k and n relatively prime. Find $k + m + n$. (Purple Comet 2013)



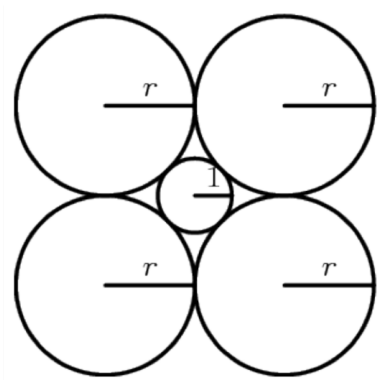
4. Equilateral triangle T is inscribed in circle A, which has radius 10. Circle B with radius 3 is internally tangent to circle A at one vertex of T. Circles C and D, both with radius 2, are internally tangent to circle A at the other two vertices of T. Circles B, C, and D are all externally tangent to circle E, which has radius $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$. (AIME II 2009)



5. The length of a common internal tangent to two circles is 7, and a common external tangent is 11. Compute the product of the radii of the two circles. (ARML 1983)

6. Square ABCD has side length 2. A semicircle with diameter AB is constructed inside the circle, and the tangent from C intersects side AD at E. What is the length of CE? (AMC 10 2004)

7. A circle of radius 1 is surrounded by 4 circles of radius r. What is r? (AMC 10 2007)



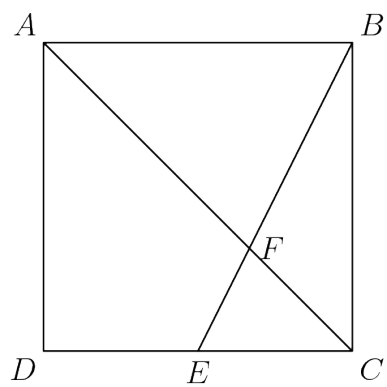
8. Circles with centers O and P have radii 2 and 4, respectively, and are externally tangent. Points A and B on the circle with center O and points C and D on the circle with center P are such that AD and BC are common external tangents to the circles. What is the area of the concave hexagon AOBCPD? (AMC 10 2006) *Hint: Similar triangles.*

9. Three circles of radius 1 are externally tangent to each other and internally tangent to circle w. Compute the radius of w. (AMC 10 2004)

10. Points A and B lie on a circle centered at O, and AOB = 60 degrees. A second circle is internally tangent to the first and tangent to both OA and OB. What is the ratio of the area of the smaller circle to that of the larger circle? (AMC 10 2008)

11. A triangle has vertices $A(0,0)$, $B(12,0)$, and $C(8,10)$. The probability that a randomly chosen point inside the triangle is closer to vertex B than to either vertex A or vertex C can be written as $\frac{p}{q}$, where p and q are relatively prime positive integers. Find $p+q$. (AIME II 2017)

12. Point E is the midpoint of side $\overline{CD}$ in square $ABCD$, and $\overline{BE}$ meets diagonal $\overline{AC}$ at F. The area of quadrilateral $AFED$ is 45. What is the area of $ABCD$?



13. Three mutually tangent spheres of radius 1 lie on a horizontal plane. A sphere of radius 2 rests on them. Compute the distance from the plane to the top of the larger sphere. (AMC 10 2004)

14.

15. DEB is a chord of a circle such that $DE = 3$ and $EB = 5$. Let O be the center of the circle. Extend OE to the circle such that the ray OE intersects the circle at C. If $EC = 1$, what is the radius of the circle? (Canada 1971)